



WEXFORD COUNTY, MICHIGAN

Landfill Fact Sheet

July 2006

INTRODUCTION

Work to address groundwater problems at the landfill has been on-going since impacts were first detected north of the landfill in the late 1980's. The latest round of investigative work, which began in 2003, has reached a significant milestone in the cleanup process. The milestone, completion of the Draft Remedial Investigation/Feasibility Study report, happens after a problem is sufficiently defined and potential cleanup options are analyzed.

This Landfill Fact Sheet has been prepared to update the community on the findings from the work completed, the potential remedies that the County is evaluating, and actions that will occur in the near future to provide residents with a clean supply of drinking water.

THE REMEDIAL ACTION PLAN PROCESS

The Remedial Action Plan (RAP) process has been developed by Michigan Department of Environmental Quality (MDEQ) to investigate contamination problems, select a remedy that is protective of human health and the environment, and then oversee the implementation of the selected remedial alternative. The RAP process includes three primary activities:

- Remedial Investigation (RI) to delineate the types of contaminants present, delineate the extent of the impacted area, and identify the contaminant source(s),
- Feasibility Study (FS) to identify and evaluate potential remedies based on the findings of the RI and other engineering factors, and
- Remedial Action Plan to select and implement the remedy.

The RAP process also includes public meetings to inform the community of the work completed and solicit public input during the remedy selection process.

Remedial Investigation

During completion of an RI, information is gathered from the site by drilling and sampling. The data gathered is then interpreted to decide if the problem is delineated. If more information is needed, additional drilling and sampling occurs until sufficient

data is present to complete the RI report and the FS report. This data gathering and interpretation phase has been on-going since 2003. To help speed the process, monthly meetings have been held with the MDEQ since last summer. Investigation activities completed include:

- Monitoring Well Installation
- Vertical Aquifer Sample (VAS) profiling
- Aquifer Characterization (Slug Tests)
- Clay Aquitard Borings
- Groundwater Analyses
- Groundwater Transport Modeling
- Data Evaluation and Interpretation
- Draft RI Report

The Wexford County DPW submitted the Draft RI Report to the MDEQ for their review on June 20, 2006. Based on recent meetings with MDEQ, additional data collection activities will continue during the planned two-month MDEQ review period.

Feasibility Study

The FS provides a systematic means to identify, screen, and evaluate potential remedial alternatives for addressing the risk associated with the groundwater contamination. The FS is not a final decision document; the selection of a remedy comes in the RAP document discussed below. The Draft FS Report was submitted to the MDEQ for their review on June 20, 2006.

Remedial Action Plan

The final decision regarding which remedy will be put in place for the site is defined in the RAP. The RAP is prepared in accordance with MDEQ requirements and summarizes the RI, the FS, and provides detailed information on the proposed remedy, such as how it will be implemented and how it will be monitored for effectiveness. The RAP is reviewed by the MDEQ in the Cadillac District Office, and then by the MDEQ in Lansing. If the RAP is approved in Lansing, the remedy is put in place at the site.

US EPA RECORDS CENTER REGION 5



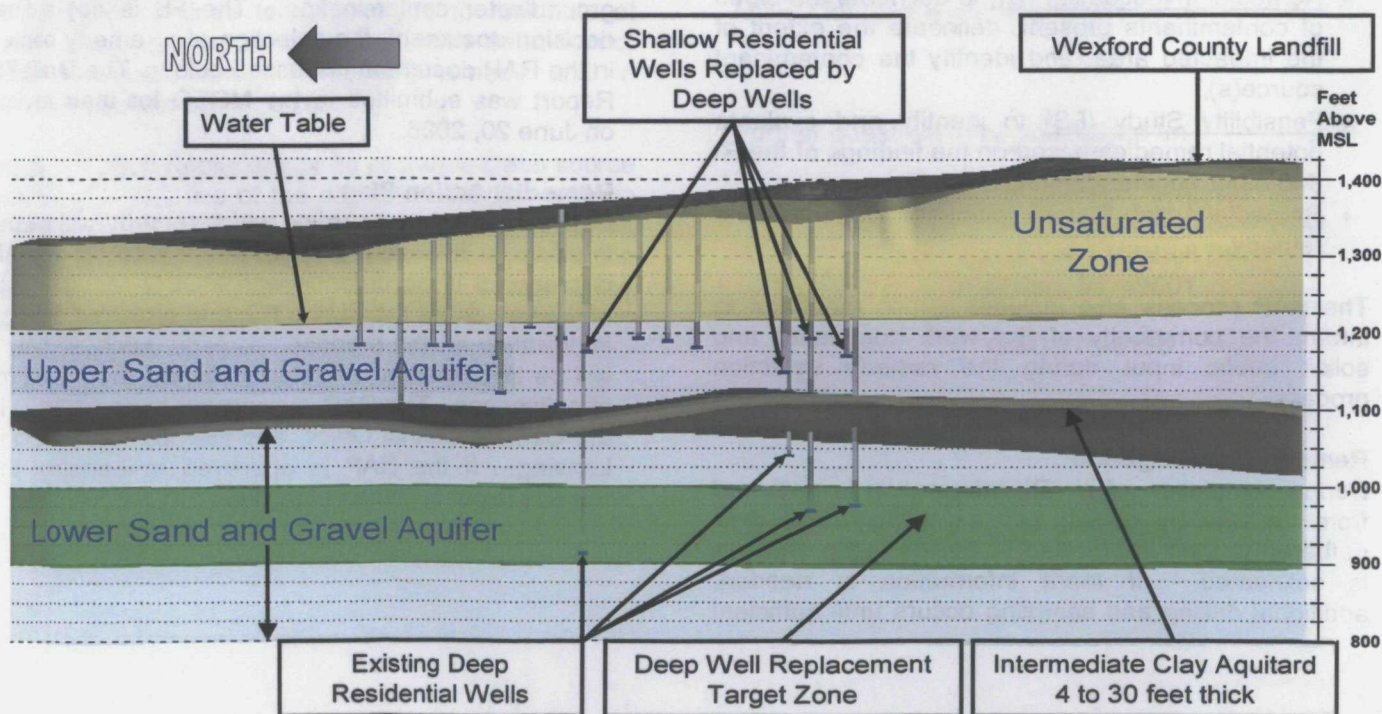
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REMEDIAL INVESTIGATION FINDINGS

The RI Report documents the hydrogeology (the setting in which groundwater flow occurs) and the extent of contamination in the investigation area (what contaminants are present and where are they located). The RI Report findings include the following:

- The area is characterized by three hydrogeologic units including the Upper Sand and Gravel Aquifer, an Intermediate Clay Aquitard, and the Lower Sand and Gravel Aquifer. The figure below depicts the three hydrogeologic units.
- The **Upper Sand and Gravel Aquifer** is the first groundwater-bearing horizon encountered below the ground surface. Most residential wells in the area are screened in this aquifer. Groundwater flow is toward the north-northwest at a rate of 280 to 360 feet per year. The flow is primarily horizontal in the investigation area. Groundwater contaminants are present in this aquifer.
- The **Intermediate Clay Aquitard** is located beneath the upper sand and gravel aquifer and acts as a barrier to the movement of groundwater and contaminants between the upper and lower sand and gravel aquifers. The clay ranges from 4 to over 30 feet thick, extends across the entire investigation area and is 100,000 times less permeable than the sand and gravel aquifers above and below it.
- The **Lower Sand and Gravel Aquifer** is greater than 190 feet thick. At total of six residential wells draw water from this aquifer. The oldest of these well has been sampled since the mid-1990s. No site contaminants have been detected in this aquifer. This aquifer has not been extensively investigated; additional investigation is being planned.

- Groundwater sample results have identified two distinctly separate contaminant plumes, a **West Plume** coming from the landfill, and an **East Plume** coming from an unknown source located east of the landfill. The figure on the following page depicts the location of the plumes.
- In the **West Plume**, several volatile organic compounds (VOCs) have been identified. The contaminants exhibit a significant downward trend in concentration, and in the number of contaminants detected since 2003. Iron and manganese are present. The highest concentrations of organic contaminants in the west plume are approximately 1,200 feet north of the site.
- In the **East Plume**, trichloroethene (TCE) is the only organic contaminant detected. TCE exhibits intermittent and/or consistently low concentrations. Iron and manganese are present. The highest concentrations of TCE in the east plume are approximately 3,000 feet north of the site.
- The horizontal extent of the west and east plumes is currently delineated by a network of monitoring and residential water supply wells. The west and east plumes extend vertically to the base of the Upper Sand and Gravel Aquifer.
- Concentrations of organic contaminants observed below the landfill are lower than those observed to the north of the landfill.
- A **Primary Zone** (where contaminants are currently located) and a **Secondary Zone** (where contaminants may migrate in the future) have been established.



FEASIBILITY STUDY REPORT FINDINGS

The FS provides a systematic process to identify, screen, and evaluate applicable remedial alternatives for addressing the human health risk associated with the groundwater contamination. The FS process includes:

- Screening available cleanup technologies,
- Assembling the applicable technologies into remedial alternatives,
- Analyzing the remedial alternatives individually,
- Comparing the remedial alternatives to each other,
- Identifying additional data needs, as appropriate,
- Identifying a preferred remedial alternative.

Several potential remedial technologies were screened. Four applicable technologies were evaluated in detail as follows:

- **Continued Interim Action** – Continue to support existing residential well treatment systems and bottled water, and provide treatment systems and bottled water and deep well replacements to future residences;
- **Point Source Treatment** – Provide well treatment systems to residents and sample treatment systems annually;
- **Deep Well Replacement** – Replacement of shallow wells for existing residences and installation of deep wells for future residences. All shallow wells will be abandoned. Deep wells will be sampled annually.
- **Public Water Supply** – Engineering, construction, oversight, and permitting of a public water supply. All existing residential wells would be abandoned and all residences hooked-up to public water supply.

By rating each remedial action according to the FS screening criteria, an overall rating has been established as shown below:

Criteria	Continue Interim Action	Point Source Treatment	Deep Well Replacement	Public Water Supply
Overall Protection of Human Health and the Environment	Medium	Medium	Medium High	High
Compliance with Applicable/Relevant Law	High	High	High	High
Reduction in Toxicity, Mobility, or Volume	Medium Low	Medium	Low	Low
Short-Term Effectiveness	High	Medium High	Medium High	Medium
Long-Term Effectiveness	Medium	Medium High	Medium High	High
Implementability	High	Medium High	Medium High	Low
State Acceptance	Low	Low	Medium	High
Community Acceptance	Medium Low	Medium Low	Medium Low	Medium Low
Cost	\$1.3M	\$2.5M	\$1.3M	\$3.5M
Overall Rating	Medium	Medium	Medium High	Medium

Based on the evaluation of remedial alternatives in the FS, the following conclusions can be made:

- Each of the four alternatives could be used to control the ingestion of contaminated groundwater.
- Continued Interim Action, Point Source Treatment, and Public Water Supply ranked **Medium**.
- Deep Well Replacement ranked **Medium-High**.



INTERIM REMEDIAL ACTIONS COMPLETED

Wexford County DPW has completed several interim remedial actions since the presence of VOCs were first detected in the monitoring and residential wells. These actions were undertaken to protect human health of residents in the area of impacted groundwater and to prevent the on-going release of contaminants from the landfill to the environment

Residential Well Sampling: – Sampling of one residential well in the mid-1990s led to replacement of that well, and the County's long-term residential well sampling program. Since 2004, approximately 40 additional residential wells along US 131, 22 Road and 41 ½ Road have been sampled. Residential wells are sampled on a quarterly basis. The location and number of wells in the quarterly residential sampling program has varied over time based on the results from the wells and the on-going site investigation.

Wells to the south, east, and west of the landfill are not in the path of groundwater flow from the landfill and are not being sampled. Wells located along 41 ½ Road were previously sampled, but are now not considered to be in the groundwater flow path from the landfill and are not being sampled.

The residential sampling program is a critical part of the County's efforts to protect the drinking water of its residents and has been very helpful in defining the extent of groundwater impacts north of the landfill.

Bottled Water and Carbon Filtration Systems – There are currently residences in the impacted area that have carbon filtration systems to remove the VOCs from the water drawn from their well. Residences are also currently being supplied with bottled water for drinking or cooking. Additional residences will be supplied with bottled water in the near future. Filtration systems and bottled water continue to be provided to residents at homes where detectable chemicals have been found. These measures are a temporary remedy until a permanent solution is in place.

Residential Well Replacement To provide a clean source of drinking water, five of the original residential wells in the area of impacted groundwater have been replaced with deeper wells. The deep wells are screened in the Lower Sand and Gravel Aquifer. These deep replacement wells are included in the quarterly

residential well sampling program to verify that the water remains free of contaminants.

Landfill Waste Relocation – In 2005, a three phase waste excavation and relocation project was initiated to remove waste from the unlined landfill areas to lined landfill cells. Total waste excavation and relocation at the Wexford County Landfill is expected to exceed 1.8 million cubic yards. The complete removal of waste from the unlined areas will substantially reduce the potential for the release of contaminants from the landfill in the future.

Groundwater Treatment System – The groundwater treatment system is located along the landfill's north property boundary and has operated since 1994. The system pumps up to 750,000 gallons per day of groundwater from the Upper Sand and Gravel Aquifer, treats the water on-site in an aeration pond to remove the organic contaminants, and returns the water to the aquifer. The future operation of this system will be incorporated into the RAP for the site.

FUTURE ACTIVITIES SCHEDULE

August 3rd, 2006: Public Meeting at the Manton Consolidated High School at 7:00 PM to discuss the Draft RI and FS reports.

Please return any questions you may have for the August meeting on the addressed post card so they can be addressed at the meeting.

September 18th, 2006: A second public meeting has been tentatively scheduled. By this date it is anticipated all remaining drilling at the site will be completed and that the County will have received the MDEQ's RI/FS review comments.

October 10th, 2006: Submittal of the Draft RAP to MDEQ for review and approval.

Further questions may be directed to:

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